

Tirthankar Roy *Software Developer*

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SUMMARY

Full-stack developer with 2.5+ years of experience building scalable applications using Node.js and Python, delivering robust backend services and efficient data-driven solutions. I've developed high-performance APIs and optimized databases to improve system reliability and response times. My biggest achievement has been driving end-to-end backend features that streamlined workflows and handled complex business logic. I'm motivated by solving challenging problems, continuously learning, and building systems that scale seamlessly.

PROFESSIONAL EXPERIENCE

21K School, Full Stack Developer - Built and optimized ERP reporting modules using complex SQL queries, improving report generation speed and reliability - Developed the Summer Camp feature in Learning Garage, enabling seamless course purchases without affecting existing workflows - Led backend development for Learning Floww, designing scalable APIs and robust system architecture - Implemented slot conflict resolution logic to handle overlapping schedules and improve booking accuracy - Integrated external APIs and optimized databases to ensure real-time data consistency and high-performance backend operations	03/2026 – Present Bengaluru
Wemotive Technologies, Software Developer - Built backend from scratch for a University of Miami research project, handling data modeling and APIs, improving response time by 50% - Developed key features for a hospital management system, including scheduling and availability, optimizing APIs by 20–30% - Designed and implemented advanced payment workflows for a cab platform, enhancing user experience and system reliability - Contributed to ONDC logistics integration, building a scalable Order Management System for complex, multi-step workflows - Developed RESTful APIs, managed async operations and integrations, and deployed production-ready applications using Docker	04/2023 – 02/2026 Pune

EDUCATION

Bachelor Of Technology, Information Technology, Central Institute of Technology	07/2017 – 08/2021 Kokrajhar, Inida
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SKILLS

Python | Fast API | Javascript | Typescript | NoSQL | mYSQL | Node Js | EC2 | RabbitMQ | DSA | React | Agile | Linux | Docker | Microservices | S3 | REST API | Sequelize | TypeORM

LANGUAGES

English | Hindi | Bengali | Assamese

PROJECTS

Coupon Management System  - Built a scalable coupon engine using Node.js and TypeScript, supporting flexible discount types like cart-wise, product-wise, and BxGy - Implemented robust validation and precise pricing logic with strong Jest test coverage to ensure accuracy and reliability	10/2025 – 11/2025
Medicine Reminder & Refill Management System  - Built a production-grade FastAPI backend for medicine tracking and refill prediction with secure JWT auth and optimized MySQL performance - Implemented automated stock monitoring with real-time alerts, ensuring timely refills and improving system reliability	11/2025 – 12/2025
E-Commerce Website - Developed scalable backend APIs for an Amazon-like e-commerce platform, managing users, products, carts, and orders - Implemented secure JWT authentication, encrypted password storage with Bcrypt, and integrated payment and shipping services with robust validation and error handling	05/2023 – 06/2023

School Management System

12/2025 – 01/2026

- Designed and developed scalable RESTful APIs using Node.js, TypeScript, and Express for managing students, enrollments, subjects, and learning goals
- Implemented secure JWT authentication, role-based access control, and optimized PostgreSQL data models using Sequelize ORM

Personal Exposure Reporter

09/2024 – 10/2025

- Built the complete backend architecture for a University of Miami project, handling database design, ERD creation, scalable APIs, and complex business logic end-to-end
- Implemented secure RBAC authorization and optimized critical SQL queries to ensure high performance, data integrity, and reliable access control

Learning Floww

04/2025 – 05/2025

- Led backend development as Dev Lead for Learning Floww, designing the complete ERD, database schema, and scalable backend architecture for the 21K School enrollment platform
- Implemented critical backend workflows including payment gateway integration, invoice generation, and unique enrollment ID creation after successful registration payments